

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_142402 | Base: main-v1.0

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Decision: major | Confidence: 78%

CHANGE REQUEST

eliminare il pannello solare di destraeliminare il pannello solare di destra

CASCADE ISSUES IDENTIFIED

[MAJOR] ISSUE-001 — Power margin reduction

Subsystem: power | Req: L1-PWR-001 | Type: constraint_violation

Removing the right solar panel reduces power generation capacity, potentially exceeding peak power consumption limits during imaging mode.

[MINOR] ISSUE-002 — Mass margin increase

Subsystem: mass | Req: L1-MAS-001 | Type: reanalysis_required

Removing the right solar panel reduces total payload mass, increasing the mass margin but potentially affecting structural dynamics.

[MINOR] ISSUE-003 — Thermal margin impact

Subsystem: thermal | Req: L1-THM-001 | Type: missing_test

Removing the right solar panel may affect thermal balance and stability, potentially impacting ΔT across orbital cycles.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-31C §5.4 — Thermal balance test to validate TMM with right solar panel removed

Reason: To ensure thermal stability after removing the right solar panel

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	53%	WARN	Thermal margin impact	■ MODIFIED
mechanical	88%	OK	1st mode 112 Hz > 80 Hz req	—
structure	91%	OK	Mass 33.8 kg / budget 35 kg	—
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-PWR-001 — verification evidence invalidated, re-analysis required
- L1-MAS-001 — verification evidence invalidated, re-analysis required
- L1-THM-001 — verification evidence invalidated, re-analysis required